

Speech Signal Processing Lab

DONG Yunyang
dongyy@sustech.edu.cn
411, No. 2, Hui Yuan

Course Arrangement

- Part 1: Lecture
- Part 2: Lab
 - Weeks 1-9,11, MATLAB exercise + knowledge sharing (start from week4)
 - Weeks 12-14, Lab project + knowledge sharing
 - Weeks 15, Lab project presentation
 - Knowledge sharing: review a topic on speech processing and application, and present the review
 - Project: choose a project on speech processing and application

Assessment

- **10% Attendance (Check attendance, knowledge sharing)**
- **50%** Lecture assignment and **Lab report**
- 20% Course final report
- **20% Lab project**

Lab Report

- Report should include:
 - Title
 - Name & SID
 - Introduction (optional)
 - Problem 1
 - Problem
 - Key Code Segment
 - Results & Analysis
 - Conclusion (optional)
- Submit lab report and program code (.m, .wav, Praat, etc.) in Blackboard system.
- Deadline: next Friday (No late submission)
- Report Filename: LabX_XXXXXXXX_Name.pdf
- Code Filename: LabCodeX_XXXXXXXX_Name.zip
- Example: Lab1_12312312_张三

10-min knowledge sharing

- News, knowledge, discussion on speech technology
- What is the impact?
- What is your thinking (e.g., problem, relation with your own study)?
- What is your idea to further development (e.g., methodology, application scenario)?

Lab Projects

1. Fundamental frequency monitoring
 - Real-time voice recording and F0 display via a GUI
2. Evaluation of English phoneme pronunciation
 - Record pronounced phoneme/speech, analyze, and provide rating/feedback for the pronunciation
3. Voice activity detection
 - Real-time voice recording and display VAD.
 - Test in noise background
4. Voice gender classification
5. Voice speaker identification
 - Register a target voice, and identify the ID of test voices
6. Other self-proposed topics

Lab I

MATLAB and DSP Tools Review

Data Types

- `matrix = [1 0; 0 1];`
- `num = 5;`
- `vector = [1 2 3];`
- `char = 'John';`
- `string = "John";`
- `cell = {'matrix', matrix, det(matrix)};`
- `struct.name = 'John'; struct. score = 100;`

Functions

```
function [O1,O2,...,Ok] = function_name(I1,I2,...,Im,varargin)
% ...
% function body goes here
% ...
```

- varargin - variable-length input argument list (cell)
- nargin – Number of function input arguments (integer)
- anonymous function

```
sqr = @(x) x.^a; %%define an anonymous function
```

Graphics

- figure
- xlabel
- title
- hold on
- grid on
- subplot
- xlim
- axis
- plot
- strips
- scatter
- bar

Audio Operation

Audio Toolbox

Read or Write Audio

<code>audioread</code>	读取音频文件
<code>audiowrite</code>	写音频文件

```
>> [s,fs] =audioread('mhint_01_01.wav');  
>> sound(s, fs);
```

Play Audio

<code>sound</code>	将信号数据矩阵转换为声音
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Object for Audio

<code>audioplayer</code>	用于播放音频的对象
<code>play</code>	从 <code>audioplayer</code> 对象播放音频
<code>playblocking</code>	从 <code>audioplayer</code> 对象播放音频，保持控制权直到播放完成

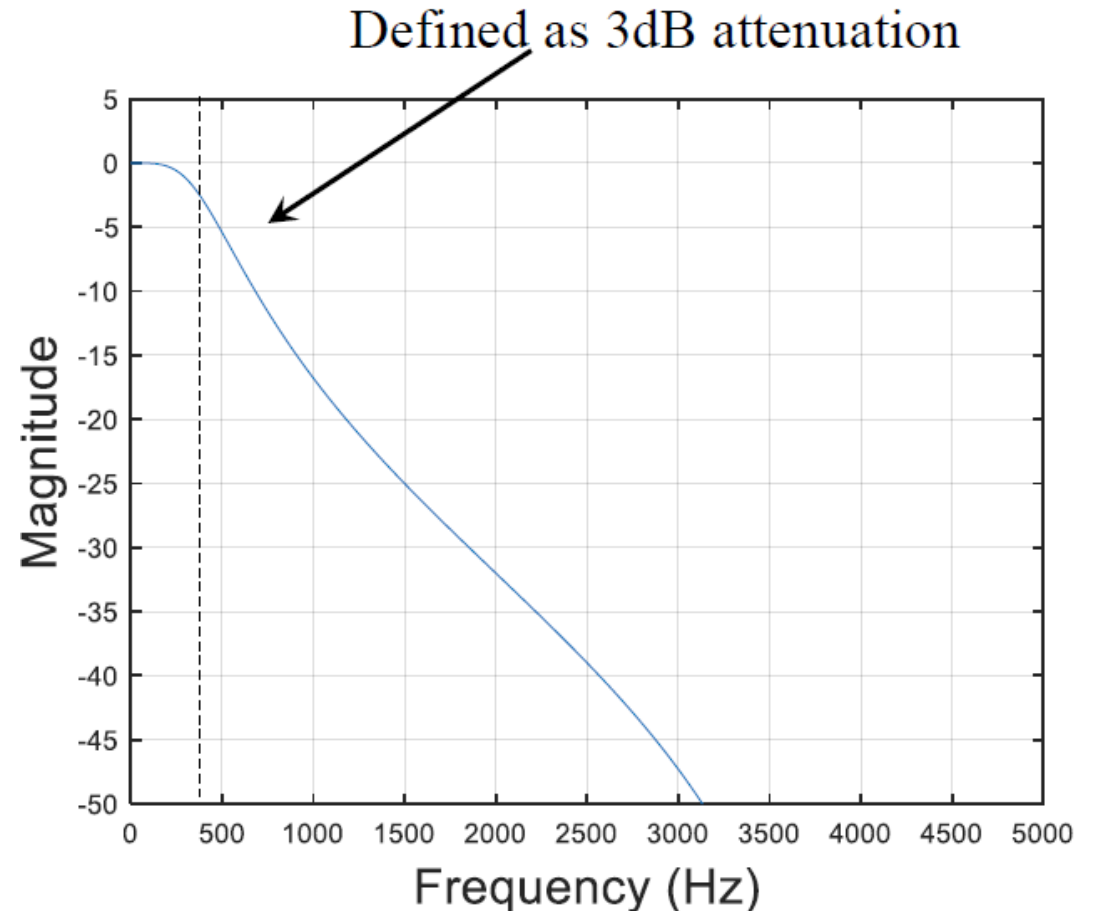
```
>> player = audioplayer(s, fs);  
>> player.playblocking();
```

Filter

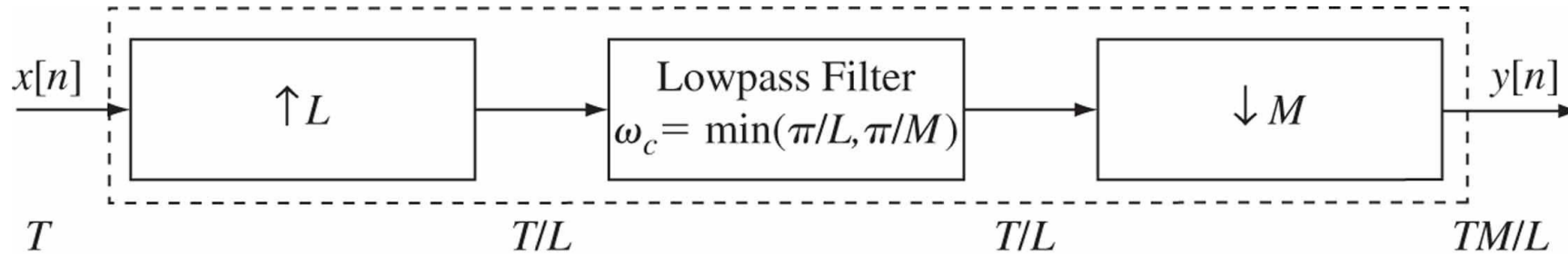
```
>> fs=8000;
%% sampling rate for discrete signal
>> fcut=400;
>> [b, a]=butter(2, fcut/(fs/2));
>> [h,f]=freqz(b, a, 512, fs);
%% Digital filter frequency response
>> plot(f, mag2db(abs(h)));
%% in dB scale
>> axis([0 5000 -50 5]);
>> xlabel('Frequency (Hz)', 'fontsize', 18)
>> ylabel('Magnitude', 'fontsize', 18)
>> grid on
```

`[b,a] = butter(n, Wn, 'ftype')`

%n: 滤波器阶数, Wn: 截止频率对 $f_s/2$ 归一化



Resample



`y = resample(x, p, q);`
resamples the input sequence x , at p/q times the original sample rate. p, q : positive integers
`[p,q] = rat(f2, f1); % from f1 to f2`

```
[s,fs]=audioread('mhint_01_01.wav');  
[P,Q]=rat(20000/fs);  
rs_20k=resample(s,P,Q);
```

`xq = interp1(n, x, nq);`
Vector n contains the sample points, and x contains the corresponding values, $x[n]$. Vector nq contains the coordinates of the query points.

Problem 1

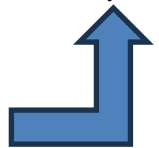
2.32. (MATLAB Exercise) Write a MATLAB program to do the following: `function load_play_plot(...)`

- Accept an arbitrary speech filename of a speech file in .wav format, i.e., filename .wav, and load the file using the MATLAB ~~wavread~~ command). `audioread`
- Listen to the speech file using the MATLAB `sound` command. (Pay attention to the required scaling for the speech file.) Write down the sentence you heard. `sound`
- Plot N sample values of the speech file (using appropriate MATLAB plot commands), starting at sample $fstart$ and ending with sample $fstart+N-1$. Plot M samples of speech per line, with up to four lines of plotting per page. The plotting parameters $fstart$, M , and N should all be specified at run time (i.e., they should be read into the program).

Use the file ~~s5.wav~~ that is available on the book website as your speech test file, and use a value of $fstart=2000$ and values of M and N that are equivalent to plotting 100 msec of speech per line and a total of $N=22000$ samples of speech being plotted. (It should be clear that the values chosen for M and N depend on the sampling rate, F_s , of the speech file.)

2.33. (MATLAB Exercise) Write a MATLAB program to plot the speech file as used in Problem 2.32 with the same parameter values for starting sample $fstart$, number of samples per line, M , and total number of samples plotted, N , but instead of plotting four lines of speech per page, use the supplied MATLAB function strips_modified.m to plot all the speech samples in a contiguous format and on a single page.

mhint_01_01.wav
(BB 下载)



Problem 2

- Design an order n low-pass digital Butterworth filter with normalized cutoff frequency W_n , where f_{cut} is low-pass cut frequency and f_s is sampling rate for the digital signal.
 - $n=2, 4, 6$
 - $W_n= 0.1, 0.3, 0.4$
 - Plot three frequency responses at the same W_n in plot.
- Learn to use 'fdatool.m' or 'filterDesigner.m' to design one of the filters described above.

Problem 3

- Read the speech file `mhint_01_01.wav` (what is its sampling rate?)
- Write MATLAB scripts to change the sampling rate to 20kHz and 8kHz by using `'resample.m'`.
- Plot the above three waveform in three plots, and **show the differences clearly**.
- Repeat the above rate modulation and plotting by using `'interp1.m'`.

Problem 4

- Read the speech file `mhint_01_01.wav`
- Process the signal with low-pass filtering (`fcut=1000, 750, 500, 400, 250 Hz`)
`filter(b,a,x)`
- Describe how low-pass filtering affect your speech understanding.
`sound`
`plot`
- Process the signal with high-pass filtering (`fcut=1k, 2k, 3k, 4k, 5k, 6k Hz`)
- Describe how high-pass filtering affect your speech understanding.